

Patient Shortage in Histopathology: How Much Do Coverage, Similarity, and Patient Count Explain?

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Abstract

On TCGA-UT, ten random patients per cancer type train a classifier 5.03 percentage points more accurate than five random patients with the same budget of 160 patches per class. Two measured cohort properties are candidate causes: coverage, the distance from held-out patients to their nearest training patient, and between-patient similarity. In random cohorts both change together with the patient count, so their contributions cannot be told apart. This protocol varies coverage and similarity by design on a three-by-three grid of target levels at five and ten patients, alongside random cohorts of both sizes. A regression of class recall on the achieved coverage, similarity, and patient count splits the gap between the random cohorts into a coverage part, a similarity part, and a remaining patient-count part, each read against a one-point threshold. A manipulation check before training confirms in all three splits that the designed cohorts vary both properties independently. Because classes with five and ten patients share one classifier, the gap between the random cohorts widens to 9.97 points. Coverage explains 2.23 points of it (95% interval: 1.40 to 2.99), similarity at most a small part (-0.04 points), and 8.08 points (7.15 to 9.09) remain for the patient count at equal coverage and similarity. Ten patients that cover held-out patients more poorly than five random patients are still more accurate than five patients that cover them as well as ten random patients. Similarity nevertheless affects recall: applied to coverage-maximizing selection of five patients, the model predicts 5.37 of the observed 5.52-point gain and attributes about half of it to the low similarity of the selected patients.

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1 Motivation

At 160 patches per TCGA-UT class, twenty patients with eight patches each outperform five patients with 32 patches each by 9.52 percentage points of patient-macro balanced accuracy; ten patients with sixteen patches each lie in between, at 67.28 against 62.25 percent (Queisler, 2026c). Effective support, which discounts patches by their correlation within a patient, explains less than half of this benefit: each doubling of patients still adds 2.62 points at equal effective support (Queisler, 2026d).

Two cohort properties are candidate explanations for the rest. *Coverage* describes how close held-out patients lie to their nearest training patient; additional patients may supply appearances of a cancer type that a small cohort misses. *Between-patient similarity* describes whether training patients deviate from their class mean in the same feature directions; similar patients add fewer independent observations. In random cohorts, both properties are nearly fixed by the patient count. A relation fitted to such cohorts lowers the residual benefit of additional patients from 2.62 to 0.79 points per doubling, but with an interval of -1.25 to 2.82 points (Queisler, 2026a).

Changing cohort composition at a fixed patient count separates the properties from the count. Five patients chosen by coverage-maximizing selection are 5.52 points more accurate than five random patients (95% interval: 4.83 to 6.23), about as accurate as ten random patients (Queisler, 2026e). Selection, however, changes both properties at once: the mean coverage distance to validation patients falls from 0.386 to 0.324, and between-patient similarity from -0.011 to -0.094 . Matched comparisons that hold one property fixed while varying the other cannot be built for these cohorts. Cohorts matched on training patients drift apart on validation patients draw by draw, and ten patients cannot reproduce the coverage and effective support of five at the same time (Queisler, 2026b). The same assessment shows that test-patient uncertainty alone leaves every contrast with a standard error of at least 0.38 points, so practical equivalence at a one-point margin is out of reach.

This report therefore varies both properties by design instead of matching them, and asks: *how many of the 5.03 points gained from five to ten random patients does coverage explain, how many does similarity explain, and how many remain for the patient count itself?* The comparison stops at ten patients. Twenty-patient cohorts cannot be made to cover held-out patients as poorly as five random patients, so the step from ten to twenty patients cannot be varied in the same way (Queisler, 2026b).

2 Study design

Only the composition of the training cohorts changes. The design inherits the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, and multinomial logistic regression with its regularization strength selected on the validation partition. Every class receives 160 patches. The eligible pool $\mathcal{E}_{sc}$ of split s and class c contains the training patients with at least 32 patches of the class, and every cohort patient contributes the first 32 or 16 patches of the established round-robin patch order.

2.1 Two measured properties

Both properties follow Queisler (2026a). For split s and class c , let $\bar{\mathbf{x}}_i$ be patient i 's mean feature vector, $\boldsymbol{\mu}_s$ the average of these means over all classes, and $\mathbf{e}_i$ the normalized vector $(\bar{\mathbf{x}}_i - \boldsymbol{\mu}_s) / \|\bar{\mathbf{x}}_i - \boldsymbol{\mu}_s\|$. The distance between two patients is $d(i, j) = 1 - \mathbf{e}_i^\top \mathbf{e}_j$. The *coverage distance* of a cohort S is the mean distance from a reference set $\mathcal{R}$ of patients to their nearest cohort patient,

$$r(S; \mathcal{R}) = \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} \min_{j \in S} d(i, j). \quad (1)$$

Smaller values mean better coverage. Cohort search uses the eligible pool as reference set; analysis uses the class’s validation patients. Test patients enter only accuracy.

Let μ_{sc} be the average patient mean over the eligible pool and τ_{sc}^2 the between-patient variance of the class. The *between-patient similarity* of a cohort S with G patients is

$$\omega(S) = \frac{1}{G(G-1)\tau_{sc}^2} \sum_{i \in S} \sum_{j \in S \setminus \{i\}} (\bar{\mathbf{x}}_i - \mu_{sc})^\top (\bar{\mathbf{x}}_j - \mu_{sc}), \quad (2)$$

the mean correlation between the deviations of two cohort patients from their class mean. Random cohorts have ω slightly below zero, between -0.013 and -0.010 ; positive values indicate aligned deviations and negative values patients that deviate in opposite directions.

2.2 Designed and random cohorts

Coverage and similarity are varied on a grid of target levels that is placed where both patient counts can reach it (Table 1). For each split, class, and draw, $\bar{r}_G$ denotes the mean coverage distance of 2,000 random cohorts of G pool patients, measured on the eligible pool, and $\Delta' = \bar{r}_5 - \bar{r}_{10}$ the coverage gained from five to ten random patients. The coverage targets are $\bar{r}_{10}$, $\bar{r}_{10} + \Delta'/2$, and $\bar{r}_{10} + \Delta'$, so they span the range from random ten to random five patients. The similarity targets are $\tilde{\omega}_{10} - 0.04$, $\tilde{\omega}_{10}$, and $\tilde{\omega}_{10} + 0.04$, where $\tilde{\omega}_{10}$ is the median similarity of the random ten-patient cohorts. Crossing the three levels of each property gives nine designed cells, which are built at five patients with 32 patches and at ten patients with 16 patches. Two random cells, five and ten uniformly drawn pool patients, anchor the design to ordinary sampling.

Cohort	G	m	Coverage targets	Similarity targets	Cells
Designed	5	32	$\bar{r}_{10}, \bar{r}_{10} + \Delta'/2, \bar{r}_{10} + \Delta'$	$\tilde{\omega}_{10} - 0.04, \tilde{\omega}_{10}, \tilde{\omega}_{10} + 0.04$	9
Designed	10	16	$\bar{r}_{10}, \bar{r}_{10} + \Delta'/2, \bar{r}_{10} + \Delta'$	$\tilde{\omega}_{10} - 0.04, \tilde{\omega}_{10}, \tilde{\omega}_{10} + 0.04$	9
Random	5	32	–	–	1
Random	10	16	–	–	1

Table 1: The twenty cells vary coverage and similarity independently at both patient counts. G is the number of patients and m the patches per patient, so every cell uses 160 patches per class. Targets are computed per split, class, and draw from random cohorts of the eligible pool.

Every classifier is trained on one split and one draw, with each of the 30 classes assigned to one cell. Class c receives cell $(o_{sc} + d) \bmod 20$ in draw d , where the offsets $o_{sc} \in \{0, \dots, 19\}$ are drawn once per split so that each offset occurs once or twice. Over every twenty draws, each class therefore passes through all twenty cells once, and each classifier mixes cells across classes. The number of draws per split is 20, 40, or 60, fixed by the precision simulation of Section 3.4.

A designed cohort is found by searching the eligible pool for the patient set closest to its two targets. Candidates combine random cohorts, prefixes of coverage-maximizing selection, and farthest-first selection from random anchor patients; swaps of single patients then minimize $|r - t_r|/0.005 + |\omega - t_\omega|/0.01$, where t_r and t_ω are the cell’s targets (Queisler, 2026b). The search uses training geometry only and fresh random seeds. Because the analysis uses achieved rather than targeted values, cohorts that miss their targets remain in the analysis.

2.3 Manipulation check before training

The cohorts of all planned draws are built and checked before any classifier is trained. The check passes only if all four conditions hold in each of the three splits; the first, second, and fourth hold separately for five and ten patients.

1. *Spread.* Averaged over classes and draws, the achieved coverage distance on the eligible pool differs between the poorest and best coverage level by at least $0.8\Delta'$, and similarity differs between the highest and lowest level by at least 0.072.
2. *Separation.* Across all designed cohorts, the correlation between class-centred coverage distance and class-centred similarity lies within $[-0.3, 0.3]$.
3. *Overlap.* A cell is on target when coverage lies within $\Delta'/4$ and similarity within 0.02 of the targets. On average over classes and draws, at least half of the nine cells are on target at both patient counts.
4. *Transfer.* The coverage difference between the poorest and best level, measured on validation patients, is at least 0.7 times the same difference on the eligible pool.

A search with the same grid, five draws per split, and other seeds met all conditions; its narrowest margin was the five-patient spread, which reached 0.89 of Δ' in each split (Queisler, 2026b). In that search, 94 to 96 percent of cohorts reached their targets, the absolute correlation between coverage and similarity was at most 0.06, and the transfer ratio lay between 0.87 and 1.00. If the check fails in any split, the experiment stops before training with its thresholds unchanged.

3 Evaluation

The evaluation estimates how strongly class recall responds to coverage, similarity, and patient count, and then translates these responses into parts of the five-to-ten-patient gap.

3.1 Class-recall model

Let y_{fc} be the patient-macro recall of class c in fit f , in percent: patch recall is averaged within each test patient of the class and then over these patients. Averaging y_{fc} over classes gives patient-macro balanced accuracy, the endpoint of the preceding experiments. The model

$$y_{fc} = \alpha_{sc} + \alpha_f + \delta r_{fc} + \lambda \omega_{fc} + \gamma z_{fc} + \varepsilon_{fc} \quad (3)$$

is fitted by least squares to all cohorts of all twenty cells. Here r_{fc} is the validation coverage distance and ω_{fc} the similarity of the cohort, z_{fc} equals one for ten patients with sixteen patches and zero for five patients with 32 patches, α_{sc} is an intercept per split and class, and α_f an intercept per fit. The coefficient δ is the change in recall per unit of coverage distance, λ per unit of similarity, and γ the effect of ten instead of five patients at equal coverage and similarity. Class intercepts remove differences in difficulty between cancer types, and fit intercepts remove shared effects of a split and draw.

3.2 Parts of the five-to-ten-patient gap

Let $\bar{r}_{\text{ran}5}$, $\bar{r}_{\text{ran}10}$, $\bar{\omega}_{\text{ran}5}$, and $\bar{\omega}_{\text{ran}10}$ be the mean validation coverage distance and similarity of the two random cells. The gap between ten and five random patients splits into three parts:

$$C = \delta (\bar{r}_{\text{ran}10} - \bar{r}_{\text{ran}5}), \quad S = \lambda (\bar{\omega}_{\text{ran}10} - \bar{\omega}_{\text{ran}5}), \quad P = \gamma. \quad (4)$$

The *coverage part* C is the recall gained because ten random patients cover held-out patients better, the *similarity part* S the recall gained because their similarity differs, and the *patient-count part* P the recall that remains for more patients at equal coverage and similarity. All three are in percentage points, and positive values favour ten patients. As a calibration check, $C + S + P$ is compared with the observed difference between the random cells, $\Delta_{5 \rightarrow 10}$, the mean recall of the ten-patient random cell minus that of the five-patient random cell. On the breadth–depth grid, this difference is 5.03 points.

A secondary, descriptive analysis applies δ and λ to the stored coverage distances and similarities of the random and selected five-patient cohorts of Queisler (2026e). It splits their predicted accuracy difference into a coverage and a similarity part and compares the sum with the observed selection gain of 5.52 points. The selected cohorts’ similarity of -0.094 lies below the targeted similarity levels, so this analysis extrapolates beyond the design.

3.3 Uncertainty

Each of 2,000 paired bootstrap replicates assigns $\text{Dirichlet}(1, \dots, 1)$ weights to the distinct test patients (Rubin, 1981), shared across cells, fits, and classes, and resamples draws with replacement within each split, jointly for all classes of a draw. Recall, the model of Equation (3), and the parts of Equation (4) are recomputed in every replicate, and the 2.5th and 97.5th percentiles give 95% intervals. Coverage distances and similarities are held fixed. The three splits share patients and are not independent replications.

3.4 Precision before training

A simulation based on stored predictions of random and selected cohorts fixes the number of draws before any classifier of this experiment is trained (Appendix A). It selects the smallest of 20, 40, or 60 draws per split for which the median *half-width* of the 95% interval of each of C , S , and P is at most one point. The half-width is half the distance between the upper and lower bound of an interval, so an interval with a half-width of one point spans two points around its estimate. If no draw count reaches this precision, the experiment stops before training.

4 Interpretation criteria

Conclusions rest on the 95% intervals of the three parts and on the practical threshold of one percentage point used in the preceding experiments. Each part receives one of four readings (Table 2). Practical equivalence is not claimed, because the test-patient uncertainty does not allow intervals within $[-1, 1]$ to be expected.

Reading	Claim for the part	95% interval
Contributes	The property explains more than one point of the gap	Lower bound above 1
At most small	The property explains less than one point of the gap	Upper bound below 1
Counteracts	The property narrows the gap by more than one point	Upper bound below -1
Unresolved	Neither a contribution nor a small part is established	All other intervals

Table 2: Each of the coverage part C , the similarity part S , and the patient-count part P receives one reading. Intervals are in percentage points. A part that counteracts is also at most small.

The cause of the five-to-ten-patient gap counts as settled when every part is either a contribution or at most small, and the calibration interval for $C + S + P - \Delta_{5 \rightarrow 10}$ does not lie entirely outside $[-1, 1]$. If it does, the linear model misdescribes the random cohorts and the parts are not interpreted.

Coverage explains the gap. If C contributes while S and P are at most small, patients help mainly by covering held-out patients. Coverage distance would then be a label-free signal of patient shortage, and coverage-maximizing selection would address its cause rather than one of its correlates.

Similarity contributes. If S contributes, similar training patients add fewer independent observations beyond what within-patient correlation accounts for. A mitigation would then need to favour patients that deviate from the class mean in different directions, and similarity-adjusted effective support would be the corresponding signal.

Patient count remains. If P contributes, additional patients help beyond both patient-mean properties. Selection among available patients could then not replace acquiring patients, and the next candidate signal would compare the patch distributions of patients rather than their means. Contributions of several parts, for example of both C and P , are read jointly as a shared explanation.

5 Results

Coverage explains part of the advantage of ten over five random patients, similarity explains none of it, and most of it remains for the patient count. Between the two random cells, the gap is $\Delta_{5 \rightarrow 10} = 9.97$ percentage points (95% interval: 8.83 to 11.09). Of these, the coverage part is $C = 2.23$ points (1.40 to 2.99), the similarity part $S = -0.04$ points (-0.06 to -0.03), and the patient-count part $P = 8.08$ points (7.15 to 9.09). By Table 2, coverage and patient count contribute and similarity is at most a small part. The parts sum to 10.27 points, and the calibration difference of 0.30 points (-1.32 to 1.96) does not lie outside $[-1, 1]$, so the criterion of Section 4 for a settled cause is met.

5.1 Manipulation check and precision

The manipulation check ran for all 30 cancer types, three splits, and 20 draws per split before any classifier was trained, and it passes in every split. The narrowest margin is the five-patient coverage-distance spread of Section 2.3, which reaches 0.889, 0.893, and 0.899 of the mean Δ' in splits 0 through 2, above the required 0.8 (Table 3). The ten-patient spread reaches 0.92 to 0.94 of Δ' , and the similarity spread exceeds the required 0.072 at both patient counts. The correlation between class-centred coverage distance and similarity stays within $|r| \leq 0.06$, well inside the permitted $[-0.3, 0.3]$. Between 93 and 96 percent of designed cells lie on target, against the required 50 percent, and the transfer ratio lies between 0.87 and 1.03, above the required 0.7.

Split	Coverage spread / Δ'		Similarity spread		Transfer ratio	
	$G = 5$	$G = 10$	$G = 5$	$G = 10$	$G = 5$	$G = 10$
0	0.889	0.924	0.080	0.078	0.950	0.962
1	0.893	0.942	0.080	0.078	0.945	1.031
2	0.899	0.926	0.080	0.078	0.873	0.959

Table 3: The designed cohorts pass every condition of the manipulation check in every split. The coverage spread (required ≥ 0.8 of Δ') is narrower at five than at ten patients; the similarity spread (required ≥ 0.072) and the transfer ratio (required ≥ 0.7) pass with margin at both patient counts. Separation and overlap, omitted here, also pass in every split (Section 5.1).

The five-patient spread reproduces, at four times the draws, the 0.89 of Δ' found in the five-draw search of Queisler (2026b). At five patients, the search thus realizes about nine tenths of the distance between the coverage targets $\bar{r}_{10}$ and $\bar{r}_{10} + \Delta'$ of Table 1. The model of Equation (3) uses the achieved coverage distances, so this narrower spread shortens the range over which δ is estimated but does not shift the cohorts that enter it.

The precision simulation of Appendix A, run on the checked cohorts, selects 20 draws per split. Its criterion is the half-width of a 95% interval, half the distance between its upper and lower bound. At 20 draws, the median half-widths are 0.76 points for C , 0.07 for S , and 0.92 for P in both scenarios, so the interval of P is expected to span about 1.8 points. With training-draw deviations enlarged by 50 percent, they rise to 1.00, 0.10, and 1.17 points. The achieved half-widths, 0.80, 0.02, and 0.97 points, match the unenlarged simulation; for C , for example, the interval from 1.40 to 2.99 points has a half-width of 0.80.

5.2 Class recall across cells

The designed cohorts reach their coverage and similarity levels at both patient counts (Table 4). On validation patients, the best coverage level of five patients lies at a coverage distance of 0.330 and the poorest at 0.385, close to the 0.333 of ten and the 0.387 of five random patients. At ten patients, the levels lie slightly higher, at 0.334 and 0.395. Similarity takes the values -0.058 , -0.018 , and 0.022 at the three levels, whereas both random cells lie near -0.005 .

Cohort	Coverage level	r	Similarity level		
			Low	Middle	High
Designed, $G = 5$	Best	0.330	63.26	63.25	62.73
	Middle	0.353	63.30	60.18	60.84
	Poorest	0.385	60.78	59.14	59.66
Designed, $G = 10$	Best	0.334	70.88	70.53	68.52
	Middle	0.363	71.51	69.53	68.69
	Poorest	0.395	70.25	66.68	65.67
Random, $G = 5$		0.387	59.30 ($\omega = -0.006$)		
Random, $G = 10$		0.333	69.27 ($\omega = -0.005$)		

Table 4: Ten patients at the poorest coverage level are more accurate than five patients at the best level. Entries are patient-macro class recall in percent, averaged over the 90 class-split-draw cohorts of each cell. The best, middle, and poorest coverage levels target $\bar{r}_{10}$, $\bar{r}_{10} + \Delta'/2$, and $\bar{r}_{10} + \Delta'$; r is the achieved validation coverage distance averaged over the three cells of a level. The low, middle, and high similarity levels reach $\omega = -0.058$, -0.018 , and 0.022 on average (-0.053 in the poorest-coverage low-similarity cell at ten patients). Descriptive means without intervals.

Within each patient count, recall falls as coverage worsens. Averaged over similarity levels, five patients lose 3.22 points from the best to the poorest coverage level, from 63.08 to 59.86 percent, and ten patients lose 2.45 points, from 69.98 to 67.53 percent. Higher similarity also lowers recall, by 1.37 points from the low to the high level at five patients and by 3.26 points at ten patients, although the middle level is the least accurate at five patients.

The patient count separates the two halves of the table more strongly than either property. Ten patients at the poorest coverage level, whose coverage distance of 0.395 is poorer than that of five random patients, reach 67.53 percent. Five patients at the best coverage level, as close to validation patients as ten random patients, reach 63.08 percent. Every ten-patient cell is more accurate than every five-patient cell.

The random cells are less alike than on the breadth–depth grid. Five random patients reach 59.30 percent, 2.95 points below the 62.25 percent on the grid, and ten random patients reach 69.27 percent, 1.99 points above the 67.28 percent (Queisler, 2026c). On the grid, all classes of a classifier share one patient count. Here, every classifier combines classes with five and ten patients, so a class with five patients plausibly loses recall to competing classes trained on ten, and a class with ten patients gains it. The gap between the random cells therefore widens from 5.03 to 9.97 points.

5.3 Parts of the five-to-ten-patient gap

The class-recall model assigns recall to coverage, similarity, and patient count with coefficients whose intervals exclude zero. Each decrease of the coverage distance by 0.01 adds 0.42 points of recall ($\delta = -41.8$; -56.1 to -26.2), each decrease of similarity by 0.01 adds 0.33 points ($\lambda = -33.5$; -45.1 to -20.9), and ten instead of five patients add $\gamma = 8.08$ points at equal coverage and similarity. The residual standard deviation of class recall is 7.06 points.

Quantity	Estimate [95% interval]	Reading	Split		
			0	1	2
Coverage part C	2.23 [1.40, 2.99]	Contributes	2.04	2.10	2.60
Similarity part S	−0.04 [−0.06, −0.03]	At most small	0.26	−0.15	−0.41
Patient-count part P	8.08 [7.15, 9.09]	Contributes	7.52	8.24	8.50
$C + S + P$	10.27		9.81	10.19	10.68
Observed gap $\Delta_{5 \rightarrow 10}$	9.97 [8.83, 11.09]		10.25	8.57	11.08
$C + S + P - \Delta_{5 \rightarrow 10}$	0.30 [−1.32, 1.96]		−0.44	1.62	−0.40

Table 5: Coverage and patient count both contribute to the gap between ten and five random patients, and patient count contributes about four times as much. Estimates pool the three splits, with paired 95% intervals over test patients and draws, and readings follow Table 2. Split columns refit the model within one split and carry no intervals. All values are in percentage points; positive parts favour ten patients.

Coverage accounts for 22 percent of the gap between the random cells and patient count for 81 percent (Table 5). The coverage part follows from the slope δ and the 0.053 by which ten random patients cover validation patients better than five. The similarity part is negligible although similarity affects recall: the two random cells differ in similarity by only 0.001, whereas the designed levels span 0.08. Its interval lies entirely below zero but far above the threshold of -1 , so it is at most small rather than counteracting. In every split, C lies between 2.0 and 2.6 points and P between 7.5 and 8.5 points.

The calibration interval, -1.32 to 1.96 points, crosses both -1 and 1 . It meets the criterion of Section 4, which asks only that the interval not lie entirely outside $[-1, 1]$, but it does not show that the parts reproduce the gap to within one point. Two descriptive checks without intervals probe the linear, additive form. Fitting the model to the designed cohorts alone gives $\delta = -41.7$, $\lambda = -29.8$, and $\gamma = 8.13$, so the random cells do not drive the estimates. Allowing the slopes to differ between patient counts gives a coverage slope of -49.1 at five and -35.6 at ten patients and a similarity slope of -26.2 and -43.4 , both pairs inside the intervals of the pooled slopes, and lowers the residual standard deviation only from 7.06 to 7.02 points.

5.4 Selection gain at five patients

Applied to the random and selected five-patient cohorts of Queisler (2026e), the model predicts a selection gain of 5.37 points (3.82 to 6.76) against the observed 5.52 points (4.83 to 6.23),

a prediction error of -0.15 points (-1.83 to 1.40 ; Table 6). It divides the prediction almost equally: coverage contributes 2.59 points through a coverage-distance change from 0.386 to 0.324, and similarity 2.79 points through a similarity change from -0.011 to -0.094 .

Model	Predicted gain	Prediction error	Predicted part	
			Coverage	Similarity
Class-recall model	5.37 [3.82, 6.76]	-0.15 [-1.83 , 1.40]	2.59	2.79
Relation fitted to random cohorts	5.77 [4.80, 6.78]	0.25 [-0.87 , 1.43]	5.34	0.43

Table 6: Both models predict the observed selection gain of 5.52 points, but they attribute it differently. The class-recall model is Equation (3), fitted in this study; the relation fitted to random cohorts is that of Queisler (2026e). The prediction error is the predicted minus the observed gain, with 95% intervals in brackets. Predicted parts are descriptive and carry no intervals. All values are in percentage points.

This analysis extrapolates. The selected cohorts’ similarity of -0.094 lies 0.036 below the lowest designed level, so the similarity part rests on the linear form beyond the design. The relation fitted to random cohorts predicts the gain equally well but attributes 5.34 points to coverage and 0.43 to similarity (Queisler, 2026e). Its similarity coefficient comes from cohorts in which similarity barely varies; the class-recall model estimates it from designed variation.

6 Discussion and conclusion

Designed variation of coverage and similarity at five and ten patients separates what random cohorts cannot: how much of the advantage of more patients comes from covering held-out patients, from patients being less alike, and from the patient count itself. Coverage contributes about two points, similarity nothing, and the patient count about eight. The clearest evidence needs no model: ten patients that cover validation patients more poorly than five random patients are still about four and a half points more accurate than five patients that cover them as well as ten random patients. Read jointly, as Section 4 prescribes for contributions of several parts, coverage is a real but minor part of patient shortage, and additional patients help mainly through something that neither patient-mean property captures.

One qualification limits how far these numbers transfer. The gap decomposed here is 9.97 points, not the 5.03 points of the breadth–depth grid, because classes with five and ten patients compete within one classifier. The patient-count part therefore contains this competition, the loss of patch depth from 32 to sixteen patches, and the patient count itself, and the study cannot separate them. If the coverage slope transferred to classifiers with a uniform patient count, coverage would account for 2.23 of 5.03 points, but the slope is also estimated under competition.

Similarity matters for accuracy even though it explains nothing of the random gap. Over the designed range of 0.08, it moves class recall by about 2.7 points, comparable to the coverage range, but random cohorts of five and ten patients are equally dissimilar. The effect appears when cohorts are selected: coverage-maximizing selection lowers similarity well below random values, and the class-recall model attributes half of the selection gain of Queisler (2026e) to this change. The relation fitted to random cohorts, which could not observe similarity varying, assigned nearly all of that gain to coverage. Selection thus works through two properties at once, and its advantage should not be read as evidence that coverage alone limits small cohorts.

For patient shortage, selection among available patients and acquisition of more patients are not interchangeable. Selection improves both coverage and dissimilarity and can reach the accuracy of twice as many random patients, but at equal coverage and similarity, ten patients

remain about eight points ahead of five under the conditions of this study. The patient-mean summaries used here leave that advantage unexplained. Following Section 4, the next candidate signal compares the patch distributions of patients rather than their means, and a design in which every class of a classifier shares one patient count would remove the competition that inflates the patient-count part.

7 Limitations

Both properties summarize a patient by its mean embedding in one representation, so variation within a patient’s patches enters neither. Their effects are assumed to be linear and additive over the targeted range, and the model carries no information about coverage or similarity outside it. Similarity measures aligned deviations in feature space rather than statistical dependence between patients.

At a fixed budget, ten patients receive sixteen instead of 32 patches, so P combines patient count with patch depth; on TCGA-UT, depth adds 0.3 points from eight to 32 patches at twenty patients (Queisler, 2026c). Classes share one classifier, so the cohort of one class can affect the recall of another. The rotating assignment balances this influence over cells but does not remove it. It widens the gap between the random cells from the 5.03 points of the breadth–depth grid to 9.97 points (Section 5.2), so P also contains the advantage of a ten-patient class over five-patient classes in the same classifier. All results are conditional on frozen Virchow2 features, multinomial logistic regression, and the three overlapping splits. Probability quality is outside this study.

A Precision simulation

The simulation uses stored patient-level predictions of four cohorts with the same features, splits, and classifier: random cohorts of five patients with 32 patches and ten patients with sixteen patches from the breadth–depth grid (Queisler, 2026c), and random and selected five-patient cohorts (Queisler, 2026e). Each contributes fifteen split–draw fits. From these fits it takes two sources of variability for class recall: the deviation of a fit from the class’s mean recall, which represents variation over training draws, and test-patient Dirichlet weights, shared across classes and cells, which represent variation over evaluation patients.

A simulated study follows the design of Section 2.2 for a candidate draw count. It uses the achieved coverage distances and similarities of the manipulation check, extended to further draws by resampling the checked draws within split. Class recall is generated as the stored class mean plus the imposed effects of Equation (3) plus a training-draw deviation resampled from the stored fits of the same class, split, and patient count. The study then applies the model, the parts, and the bootstrap of Section 3.3 with 200 replicates. Two scenarios impose the effects: all parts equal to zero, and a coverage-dominant split of the 5.03-point gap into $C = 4$, $S = 0.5$, and $P = 0.5$.

Each scenario and draw count uses 2,000 simulated studies. A draw count qualifies when the median half-width of each part is at most one point in both scenarios. A sensitivity condition with training-draw deviations enlarged by 50 percent is reported but does not affect the choice.

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